



IT-SIMPLERA INSTITUTE
Red Team Cyber Security Internship Program

WEEK 03 ASSIGNMENT REPORT
Cybersecurity Fundamentals and Security Concepts

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1. Introduction

This report covers Week 03 of the Red Team Cyber Security Internship Program at IT-SIMPLERA Institute. It strengthens core networking and cybersecurity fundamentals through theoretical research and hands-on practice in Kali Linux, covering common network protocols, HTTP versus HTTPS security, network reconnaissance, DNS, and open source information gathering.

2. Task 1: Protocols and Port Numbers

The table below lists 15 common network protocols with their default port number, transport type, purpose, and security status.

Protocol	Port	TCP/UDP	Purpose	Secure/Insecure
FTP	21	TCP	File transfer between client and server	Insecure (cleartext credentials)
SSH	22	TCP	Secure encrypted remote login and file transfer	Secure
Telnet	23	TCP	Remote login (cleartext)	Insecure
SMTP	25	TCP	Sending email between mail servers	Insecure (unless STARTTLS used)
DNS	53	UDP/TCP	Resolves domain names to IP addresses	Insecure (unless DNSSEC/DoH used)
DHCP	67/68	UDP	Automatically assigns IP addresses to devices	Insecure
HTTP	80	TCP	Transfers web pages in plain text	Insecure
POP3	110	TCP	Retrieves email from a mail server	Insecure
IMAP	143	TCP	Retrieves and manages email on a server	Insecure
HTTPS	443	TCP	Transfers web pages using TLS encryption	Secure
SMB	445	TCP	File and printer sharing on a network	Insecure (common target for exploits)
SMTPS	465	TCP	Sending email securely over TLS	Secure
IMAPS	993	TCP	Secure encrypted email retrieval	Secure
RDP	3389	TCP	Remote desktop access to Windows systems	Insecure by default, needs hardening
MySQL	3306	TCP	Database client server connection	Insecure by default

3. Task 2: HTTP vs HTTPS Analysis

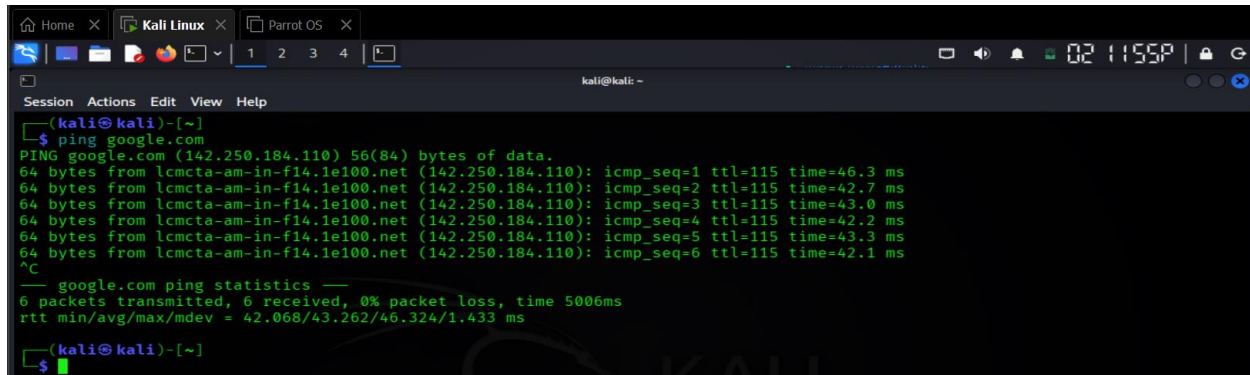
Ten websites were analyzed for their use of HTTP or HTTPS, automatic redirection, and overall connection security.

#	Website	Protocol	Auto-Redirect HTTP to HTTPS	Secure? Explanation
1	google.com	HTTPS	Yes	Enforces HTTPS with HSTS, all traffic encrypted, valid trusted certificate.
2	github.com	HTTPS	Yes	Uses HTTPS with HSTS preload, prevents downgrade to HTTP.
3	httpforever.com	HTTP	No	Purpose built to always stay on HTTP, so people can test whether a network's captive portal or login page loads before switching to a secure connection.
4	microsoft.com	HTTPS	Yes	HTTPS enforced across all pages with valid CA-issued certificate.
5	httpforever.com	HTTP	No	Built specifically so it never migrates to HTTPS, for demonstrating what plain HTTP looks like in the browser.
6	youtube.com	HTTPS	Yes	Served under Google HTTPS infrastructure with HSTS.
7	facebook.com	HTTPS	Yes	HTTPS enforced, session cookies marked secure.
8	linkedin.com	HTTPS	Yes	HTTPS with HSTS, protects login credentials in transit.
9	yahoo.com	HTTPS	Yes	HTTPS enforced, redirects any HTTP request automatically.
10	x.com	HTTPS	Yes	HTTPS enforced across the platform including media delivery.

4. Task 3: Kali Linux Network Reconnaissance

4.1 Ping - google.com

Command used: ping google.com



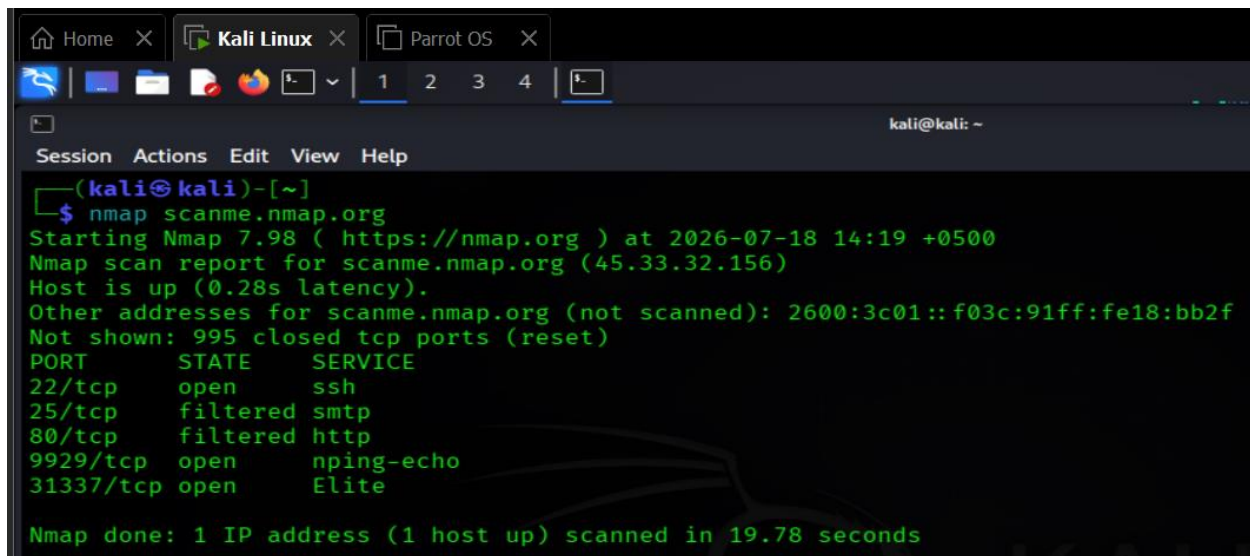
```
(kali@kali)-[~]
$ ping google.com
PING google.com (142.250.184.110) 56(84) bytes of data:
64 bytes from lcmcta-am-in-f14.1e100.net (142.250.184.110): icmp_seq=1 ttl=115 time=46.3 ms
64 bytes from lcmcta-am-in-f14.1e100.net (142.250.184.110): icmp_seq=2 ttl=115 time=42.7 ms
64 bytes from lcmcta-am-in-f14.1e100.net (142.250.184.110): icmp_seq=3 ttl=115 time=43.0 ms
64 bytes from lcmcta-am-in-f14.1e100.net (142.250.184.110): icmp_seq=4 ttl=115 time=42.2 ms
64 bytes from lcmcta-am-in-f14.1e100.net (142.250.184.110): icmp_seq=5 ttl=115 time=43.3 ms
64 bytes from lcmcta-am-in-f14.1e100.net (142.250.184.110): icmp_seq=6 ttl=115 time=42.1 ms
^C
--- google.com ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5006ms
rtt min/avg/max/mdev = 42.068/43.262/46.324/1.433 ms
(kali@kali)-[~]
$
```

Response Time (ms): around 42-46 ms

TTL Value: 115 (128-115=13 | means the packet passed through 13 routers/hops before reaching us)

4.2 Nmap Scan - scanme.nmap.org

Command used: nmap scanme.nmap.org



```
(kali@kali)-[~]
$ nmap scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-18 14:19 +0500
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.28s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
25/tcp    filtered smtp
80/tcp    filtered http
9929/tcp  open  nping-echo
31337/tcp open  Elite

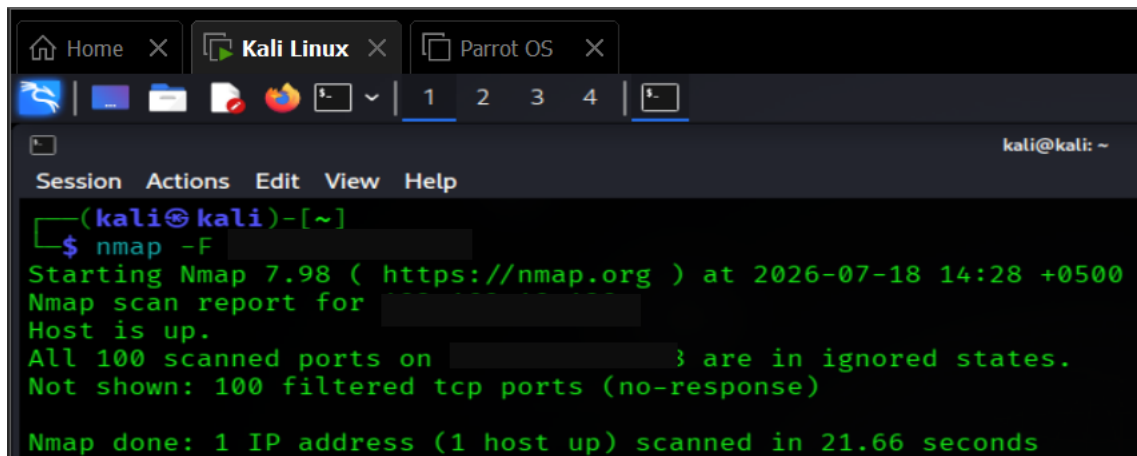
Nmap done: 1 IP address (1 host up) scanned in 19.78 seconds
```

Open Ports Found: two open ports (22/tcp, 9929/tcp, 31337/tcp)

Services Running: ssh, smtp, http, nping-echo, Elite

4.3 Nmap Scan - Local IP Address

Command used: nmap [your local IP]



```
(kali@kali)-[~]
$ nmap -F [redacted]
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-18 14:28 +0500
Nmap scan report for [redacted]
Host is up.
All 100 scanned ports on [redacted] are in ignored states.
Not shown: 100 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 21.66 seconds
```

I scanned top 100 ports by -F command

Open Ports Found: I'm scanning my VM (Kali) from the same VM itself. The Windows host's firewall (or VMware's virtual network) is silently dropping the probe packets, so nmap gets no response and marks everything "filtered."

Services Running: No running services shown as all the scanned ports are in ignored states

4.4 Security Risk Analysis

Which open port could be a potential security risk and why?

Highest Risk Port and Reason:

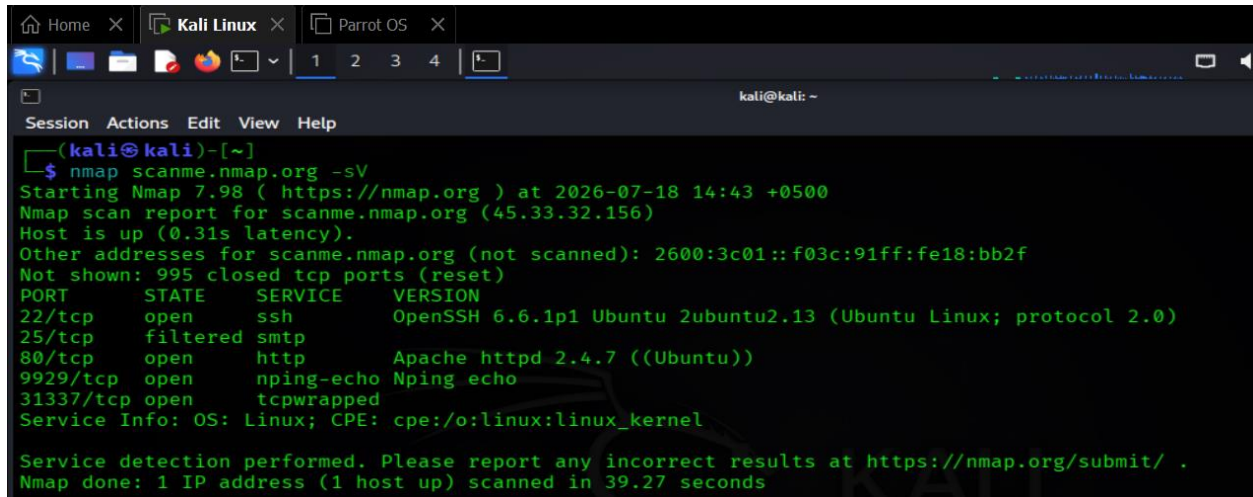
Port 445 (SMB) or **Port 3389 (RDP)** are typically the highest-risk open ports.

Port 445 (SMB): It handles file and printer sharing and has been the target of major exploits (e.g., EternalBlue, WannaCry). If exposed, it can allow remote code execution and lateral movement across a network.

Port 3389 (RDP): It provides direct remote desktop access to the system. It is frequently targeted by brute-force attacks and, if secured with weak credentials, can grant an attacker full control of the machine.

4.5 Bonus: Purpose of the -sV Option in Nmap

The -sV flag enables version detection in Nmap. Instead of only reporting that a port is open, Nmap probes the service running on that port and attempts to determine its name and version, for example **Apache httpd 2.4.7** or **OpenSSH 6.6.1**. This helps identify outdated or vulnerable software versions during a security assessment.



```
(kali㉿kali)-[~]
$ nmap scanme.nmap.org -sV
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-18 14:43 +0500
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.31s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 995 closed tcp ports (reset)
PORT      STATE      SERVICE      VERSION
22/tcp    open      ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
25/tcp    filtered  smtp
80/tcp    open      http         Apache httpd 2.4.7 ((Ubuntu))
9929/tcp  open      nping-echo   Nping echo
31337/tcp open      tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 39.27 seconds
```

5. Task 4: What Happens When You Type a URL and Press Enter

The process below is presented as a sequential stack, similar to a layered network model, showing each stage in the exact order it occurs.

	DNS Lookup
STEP 1	The browser checks its own cache, then the operating system cache, then queries a DNS resolver (usually the ISP or a public resolver such as 8.8.8.8). If not cached, the resolver queries a root server, then a TLD server, then the authoritative name server for the domain, which returns the IP address for the requested hostname.
	IP Address Resolution
STEP 2	Once the domain name is resolved, the browser now holds the numeric IP address (IPv4 or IPv6) of the destination server. This address is required so the browser knows exactly where on the network to send the connection request.
	TCP Three-Way Handshake
STEP 3	The browser initiates a TCP connection with the server using SYN, SYN-ACK, ACK. The client sends a SYN packet, the server replies with SYN-ACK, and the client confirms with ACK. This establishes a reliable connection before any data is exchanged.
	TLS/HTTPS Handshake
STEP 4	If the site uses HTTPS, a TLS handshake follows the TCP handshake. The client sends a ClientHello, the server responds with ServerHello and its certificate, the client verifies the certificate against a trusted certificate authority, and both sides agree on a shared session key used to encrypt all further communication.
	HTTP Request
STEP 5	The browser sends an HTTP request to the server, typically a GET request, including the requested path, headers such as User-Agent, cookies, and accepted content types.
	Server Response
STEP 6	The server processes the request and replies with an HTTP response containing a status code, such as 200 OK or 404 Not Found, response headers, and the requested content, usually HTML, in the response body.
	Browser Rendering the Webpage
STEP 7	The browser parses the HTML to build the DOM, parses CSS to build the CSSOM, combines them into a render tree, calculates layout, and paints pixels to the screen. Additional resources such as images, scripts, and fonts are requested and rendered as they are discovered.

6. Task 5: Networking Cheat Sheet

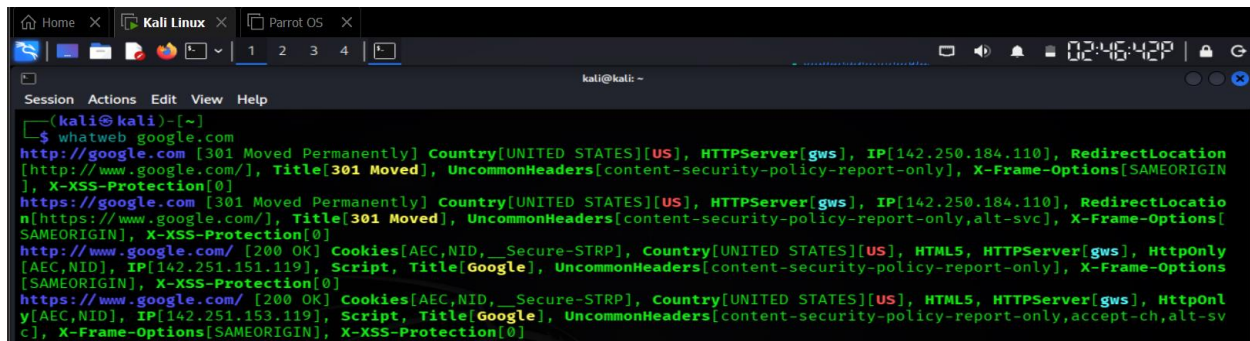
A quick reference cheat sheet of 15 common protocols, their ports, purpose, associated security risk, and a secure alternative where applicable.

Protocol	Port	TCP/UDP	Purpose	Security Risk	Secure Alternative
FTP	21	TCP	File transfer	Credentials sent in plaintext	SFTP / FTPS
SSH	22	TCP	Remote login	Weak passwords, brute force	Use key-based auth
Telnet	23	TCP	Remote login	No encryption at all	SSH
SMTP	25	TCP	Send email	Spoofing, no encryption	SMTPS (465)
DNS	53	UDP/TCP	Name resolution	Cache poisoning, spoofing	DNSSEC / DoH
DHCP	67/68	UDP	IP assignment	Rogue DHCP server attacks	DHCP snooping
HTTP	80	TCP	Web browsing	Data sent in plaintext	HTTPS
POP3	110	TCP	Email retrieval	Credentials in plaintext	POP3S (995)
IMAP	143	TCP	Email management	Credentials in plaintext	IMAPS (993)
HTTPS	443	TCP	Secure web browsing	Weak/expired certificates	Keep TLS updated
SMB	445	TCP	File sharing	Exploited by ransomware (EternalBlue)	Disable SMBv1, patch
SMTPS	465	TCP	Secure email sending	Misconfigured certificates	N/A
IMAPS	993	TCP	Secure email retrieval	N/A	N/A
RDP	3389	TCP	Remote desktop	Brute force, exposed to internet	VPN + MFA
MySQL	3306	TCP	Database access	Weak credentials, exposed port	Firewall + strong auth

7. Task 6: Kali Linux Web Enumeration

7.1 whatweb - google.com

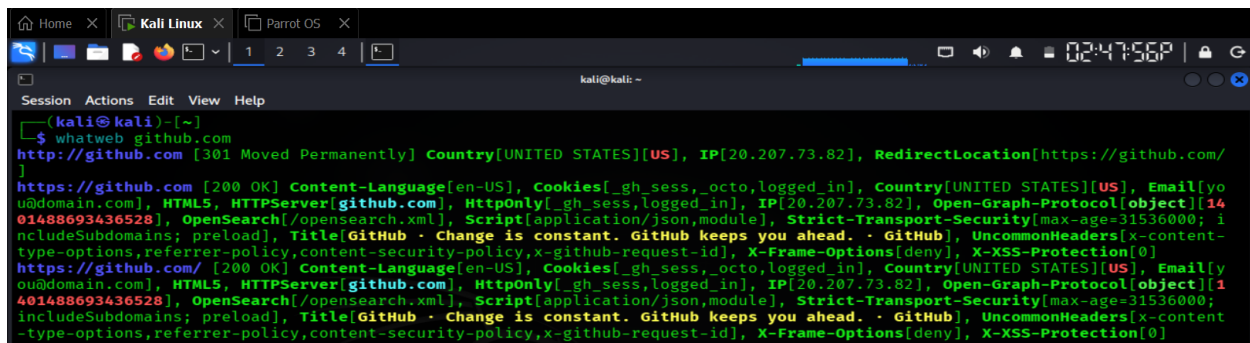
Command used: whatweb google.com



```
(kali@kali)-[~]
$ whatweb google.com
http://google.com [301 Moved Permanently] Country[UNITED STATES][US], HTTPServer[aws], IP[142.250.184.110], RedirectLocation[http://www.google.com/], Title[301 Moved], UncommonHeaders[content-security-policy-report-only], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]
https://google.com [301 Moved Permanently] Country[UNITED STATES][US], HTTPServer[aws], IP[142.250.184.110], RedirectLocation[https://www.google.com/], Title[301 Moved], UncommonHeaders[content-security-policy-report-only,alt-svc], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]
http://www.google.com/ [200 OK] Cookies[AEC,NID,__Secure-STRP], Country[UNITED STATES][US], HTML5, HTTPServer[aws], HttpOnly[AEC,NID], IP[142.251.151.119], Script, Title[Google], UncommonHeaders[content-security-policy-report-only], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]
https://www.google.com/ [200 OK] Cookies[AEC,NID,__Secure-STRP], Country[UNITED STATES][US], HTML5, HTTPServer[aws], HttpOnly[AEC,NID], IP[142.251.153.119], Script, Title[Google], UncommonHeaders[content-security-policy-report-only,accept-ch,alt-svc], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]
```

7.2 whatweb - github.com

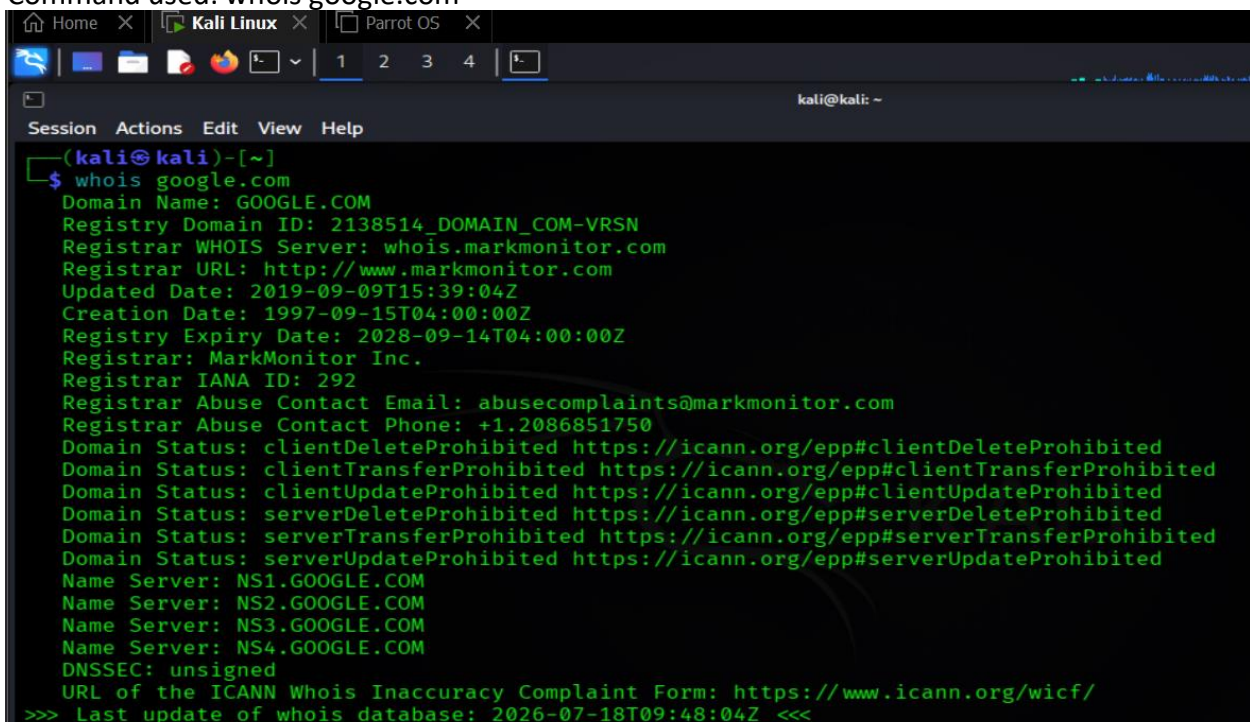
Command used: whatweb github.com



```
(kali@kali)-[~]
$ whatweb github.com
http://github.com [301 Moved Permanently] Country[UNITED STATES][US], IP[20.207.73.82], RedirectLocation[https://github.com/]
https://github.com [200 OK] Content-Language[en-US], Cookies[_gh_sess,octo,logged_in], Country[UNITED STATES][US], Email[youdomain.com], HTML5, HTTPServer[github.com], HttpOnly[_gh_sess,logged_in], IP[20.207.73.82], Open-Graph-Protocol[object][1401488693436528], OpenSearch[/opensearch.xml], Script[application/json,module], Strict-Transport-Security[max-age=31536000; includeSubdomains; preload], Title[GitHub · Change is constant. GitHub keeps you ahead. · GitHub], UncommonHeaders[x-content-type-options,referrer-policy,content-security-policy,x-github-request-id], X-Frame-Options[deny], X-XSS-Protection[0]
https://github.com/ [200 OK] Content-Language[en-US], Cookies[_gh_sess,octo,logged_in], Country[UNITED STATES][US], Email[youdomain.com], HTML5, HTTPServer[github.com], HttpOnly[_gh_sess,logged_in], IP[20.207.73.82], Open-Graph-Protocol[object][1401488693436528], OpenSearch[/opensearch.xml], Script[application/json,module], Strict-Transport-Security[max-age=31536000; includeSubdomains; preload], Title[GitHub · Change is constant. GitHub keeps you ahead. · GitHub], UncommonHeaders[x-content-type-options,referrer-policy,content-security-policy,x-github-request-id], X-Frame-Options[deny], X-XSS-Protection[0]
```

7.3 whois - google.com

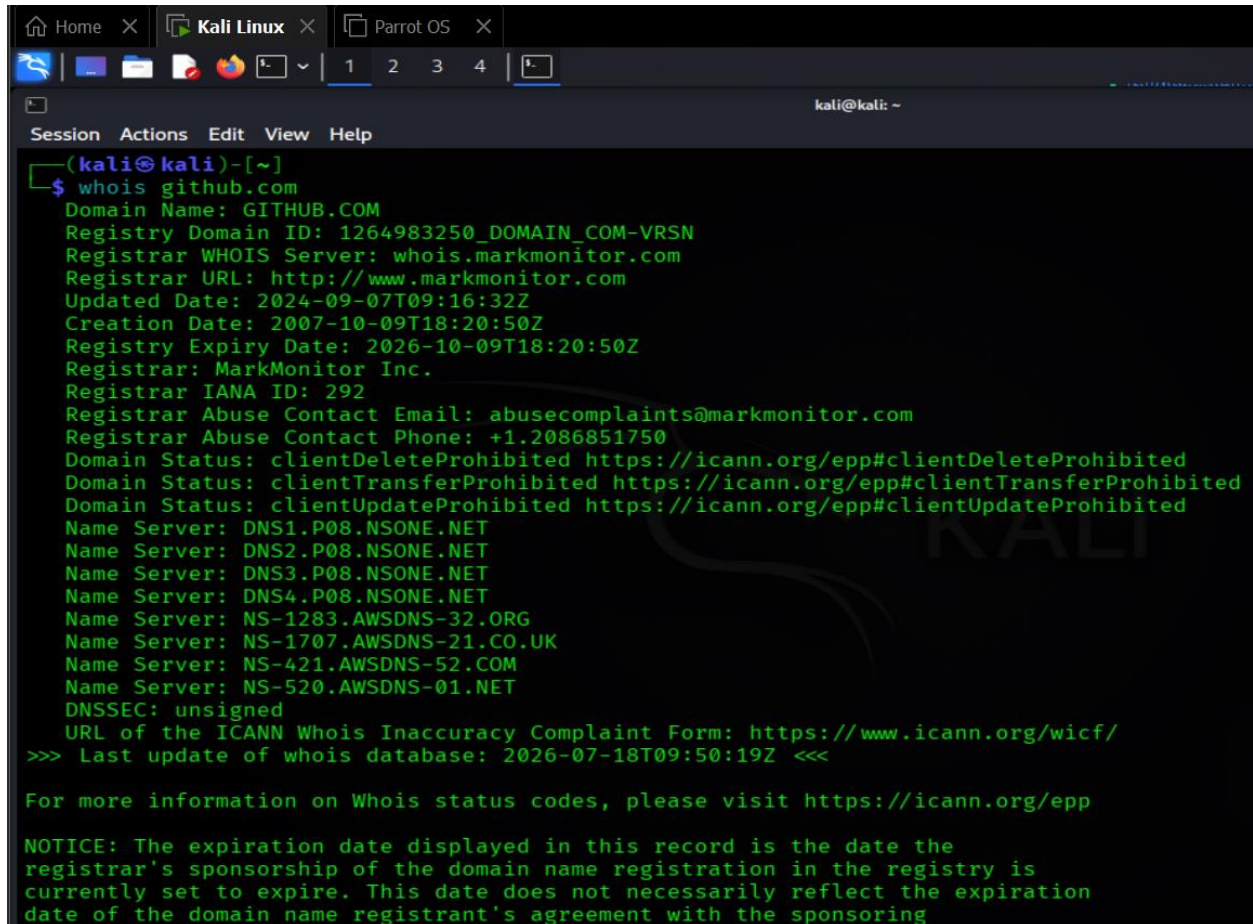
Command used: whois google.com



```
(kali@kali)-[~]
$ whois google.com
Domain Name: GOOGLE.COM
Registry Domain ID: 2138514_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2019-09-09T15:39:04Z
Creation Date: 1997-09-15T04:00:00Z
Registry Expiry Date: 2028-09-14T04:00:00Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Name Server: NS1.GOOGLE.COM
Name Server: NS2.GOOGLE.COM
Name Server: NS3.GOOGLE.COM
Name Server: NS4.GOOGLE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-07-18T09:48:04Z <<<
```

7.4 whois - github.com

Command used: whois github.com



```
(kali@kali)-[~]
$ whois github.com
Domain Name: GITHUB.COM
Registry Domain ID: 1264983250_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2024-09-07T09:16:32Z
Creation Date: 2007-10-09T18:20:50Z
Registry Expiry Date: 2026-10-09T18:20:50Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: DNS1.P08.NSONE.NET
Name Server: DNS2.P08.NSONE.NET
Name Server: DNS3.P08.NSONE.NET
Name Server: DNS4.P08.NSONE.NET
Name Server: NS-1283.AWSDNS-32.ORG
Name Server: NS-1707.AWSDNS-21.CO.UK
Name Server: NS-421.AWSDNS-52.COM
Name Server: NS-520.AWSDNS-01.NET
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-07-18T09:50:19Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
```

7.5 Summary Table

Fill in the fields below using the output captured in the screenshots above.

Field	google.com	github.com
Web Server / Technology	gws (Google Web Server), HTML5, uses security headers like CSP, X-Frame-Options, HSTS on HTTPS	github.com server, HTML5, Strict-Transport-Security (HSTS) with preload, X-Frame-Options: deny
Domain Registrar	MarkMonitor Inc.	MarkMonitor Inc.
Registration Date	1997-09-15	2007-10-09
Name Servers	ns1.google.com, ns2.google.com, ns3.google.com, ns4.google.com	dns1-4.p08.nsone.net, ns-1283.awsdns-32.org, ns-1707.awsdns-21.co.uk, ns-421.awsdns-52.com, ns-520.awsdns-01.net
Other Findings	HTTP auto-redirects to HTTPS; DNSSEC unsigned; domain expires 2028-09-14; registrant info hidden via MarkMonitor privacy form	HTTP auto-redirects to HTTPS; DNSSEC unsigned; domain expires 2026-10-09; registrant listed as GitHub, Inc.; uses AWS Route 53 + NS1 (multi-provider DNS setup)

8. Challenges Faced

Challenges faced while scanning via nmap sometimes the nmap will get stucked. Also faced problems while finding the technologies such sites like google,github is using.

9. Conclusion

This assignment reinforced core cybersecurity fundamentals, including common protocols and their associated risks, the importance of HTTPS over HTTP, practical use of Kali Linux reconnaissance tools such as ping and nmap, the complete process of loading a webpage, and open source information gathering using whatweb and whois.

10. References

- <https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml>
- <https://nmap.org/docs.html>
- <https://www.kali.org/docs/tools/>